Hierarchical Diffusion Model for Remote Sensing Image Change Detection

Pengfei Han, Yunpeng Gao, Guofang Chen, Bin Zhao, and Xuelong Li, *Fellow, IEEE*

***Abstract*—Over several years, deep learning-based methods have made significant progress in the field of remote sensing image (RSI) change detection (CD). However, challenges such as multiscale object changes and cluttered backgrounds, including seasonal variations, shadows, and vegetation color changes, often result in false positives and missed detections, which substantially affect model performance. To tackle these limitations, we present a hierarchical diffusion model for CD (HDM-CD) in RSIs, which effectively integrates spatial and frequency domain information to eliminate false positives and pseudochanges in practical applications. Specifically, we propose a hierarchical feature representation diffusion model that analyzes the complementary characteristics of multilevel information in both spatial and frequency domains. This model successfully combines the capture of local details with the perception of global structures, achieving a comprehensive representation of multiscale spatial–frequency features. Furthermore, we present an uncertainty-guided CD model that directs the model's focus on learning the features of change regions while simultaneously enhancing the network's ability to represent uncertain regions and complex boundaries. Finally, we design a spatial–frequency joint optimization module (SFJOM) to mitigate the information loss in small object change areas and accurately detect object contours and sharp edges in the change areas. Extensive experimental results indicate that the proposed HDM-CD method attains state-of-the-art (SOTA) CD performance, exceeding existing competitors by +8.21%, +44.3%, and +17.1% in intersection over union (IoU) on LEVIR-CD, DSIFN-CD, and WHU-CD dataset, respectively.**

***Index Terms*—Change detection (CD), hierarchical diffusion model, remote sensing, uncertainty.**

I. Introduction

CHANGE detection (CD) enabled within remote sensing images (RSIs) focuses on examining multitemporal images from the same geographical location to observe and track surface dynamics [1]. Being at least one of the fundamental tasks in the field of RSIs [2], CD is vital for many applications, *e.g.*, geological investigations [3], environmental monitoring [4], and agricultural evaluation [5].

The objective of the RSI-CD is to extract unchanged or changed features or information from two aligned bitemporal RSIs, resulting in the generation of a difference image, and then obtain the changed image [6], [7]. Traditional CD methods [8], [9], [10] use hand-crafted features to identify any changed or unchanged objects to generate change maps (CMs). For example, principal component analysis (PCA) [11] and multivariate alteration detection (MAD) [12] are being typically fashioned. However, these methods fail to capture high-level feature representations, leading to decreased accuracy. Hence, it is essential to develop automatic CD technologies with robust representation capabilities. Over the past few decades, learning-based approaches, particularly CNNs, have been utilized in the RSI-CD. For instance, image compression [13], object tracking [14], and salient object detection [15] of RSIs due to its superior feature representation capabilities. However, traditional convolution methods struggle in their localized aspects to capture dynamic global semantic interactions and contextual relationships among features at various levels due to their localized nature. To mitigate this limitation, newer CNN-based approaches incorporate dilated convolutions or attention mechanisms, yet they still face challenges in extracting comprehensive global information. Recently, transformer-based methods have emerged, demonstrating exceptional global context modeling capabilities and consistently achieving cutting-edge capabilities across various CD tasks.

A. Motivation and Overview

It is true that deep learning methods have brought great improvements, but there are still some weaknesses that hinder the performance to be improved further. The reasons behind this can be attributed as follows.

1) The pseudochanges arise out of climate differences in the bitime imagery. When bitime imagery has been acquired at diurnally variable times and seasons, climate of the season tends to introduce pseudochange (e.g., variation changes of shading and vegetation colors) in the detection map. Overcoming these pseudochanges resulting from phenological variations in bitemporal images remains a challenging task. As shown in the red-marked box in the 1st row of Fig. 1, the ChangeFormer

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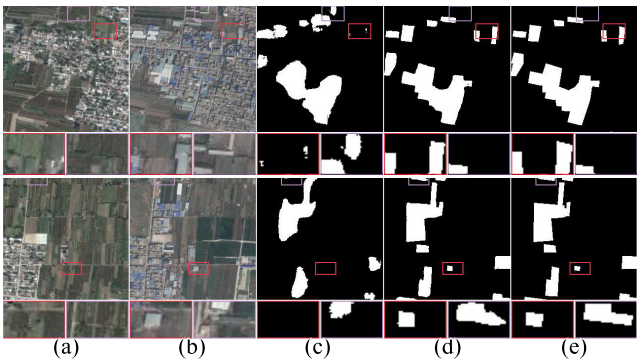


Fig. 1. Comparison of the results of the different methods. (a) Prechange image. (b) Postchange image. (c) ChangeFormer. (d) HDM-CD (ours). (e) Ground truth mask.

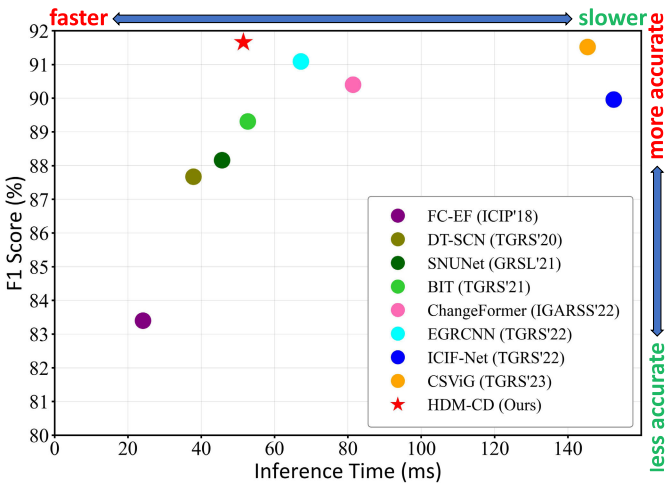


Fig. 2. Comparison of efficiency and perception across the LEVIR-CD database. The F1 score distributes on the horizontal axis, the horizontal axis represents the inference time.

identifies changed objects in the image as unchanged regions. In contrast, the proposed HDM-CD not only identifies change objects accurately, but also well keep high internal compactness and sharpness of edges of the change region.

2) Capturing various shapes and sizes of changed areas within an image scene is crucial. Intuitively, these changed areas may exhibit diverse shapes and scales, and numerous studies have shown that accurately covering these variations can boost the performance of RSIs-CD. For example, information on small and large changed objects could be grossly missinformed, to the extent that some models fail to fully isolate the background of the small objects in the red-marked box in line 3 of Fig. 1. The results indicate that the proposed HDM-CD excels in preserving both complete structure and accurate details, surpassing the accuracy of other techniques.

Based on a thorough analysis of the presented research, a hierarchical diffusion model for change detection (HDM-CD) is designed, which integrates spatial and frequency domain information through an in-depth examination of existing studies. To comprehensively analyze the complementary characteristics of multilevel information in both spatial and frequency domains, a hierarchical feature representation diffusion model is adopted, capturing local details and perceiving global structures, thereby achieving a complete representation of multiscale spatial frequency features. To address the challenges in expressing uncertain regions and complex boundaries, an uncertainty-guided CD model is introduced, focusing on learning from the zone of change. With the intention of further improving the model's robustness and detection accuracies, a spatial–frequency joint optimization module (SFJOM) is incorporated, which effectively strengthens the model's learning of desired high-frequency components, reduces information loss in small object change regions, and precisely detects object contours and change areas. The proposed model consistently outperforms prevailing best-of-breed methodologies in complex CD scenarios, demonstrating exceptional property and reliability.

B. Contributions

The proposed HDM-CD demonstrates outstanding performance in both efficiency and accuracy compared to current state-of-the-art (SOTA) technologies, as illustrated in Fig. 2. In conclusion, the key contents are summarized below.

1) We first harness a hierarchical feature representation diffusion model (HDM) that integrates the capture of local details with the perception of global structures, addressing false detection and pseudochange in real-world scenarios.
2) We introduce an uncertainty-guided model (UGM) that focuses on learning features of changed regions, enhancing representation of uncertain areas and minimizing disturbances in nonchange regions.
3) We design an SFJOM that enhances the learning of high-frequency components, ultimately optimizing model performance in detail recovery and structure retention.

II. Related Works

Leveraging the CNN's potent capabilities of feature extraction and representation, numerous CNN-inspired CD methods have attained promising results [16], [17]. Several studies have employed previously pretrained CNNs, initially developed as feature extractors intended for classification of images, to harvest deep features from raw bitime images. For example, in [18], a specialized UNet++ has been designed toward the RSI-CD task. Zhan et al. [19] describe a Siamese CNN framework which seeks to exploit features from bitemporal imagery and incorporates a contrast loss function for tackling the challenge of data imbalance. Zhang and Shi [20] leverage the VGG16 [21] model pretrained on scene-level RSI data to extract different deep features and perform controlled changes. However, these methods are often sensitive to data transfer between different domains, which may lead to adaptability issues. Daudt et al. [22] propose a comprehensive CD method that fully utilizes distributed weighted convolutional networks. To separate subject-level feature information and yield a

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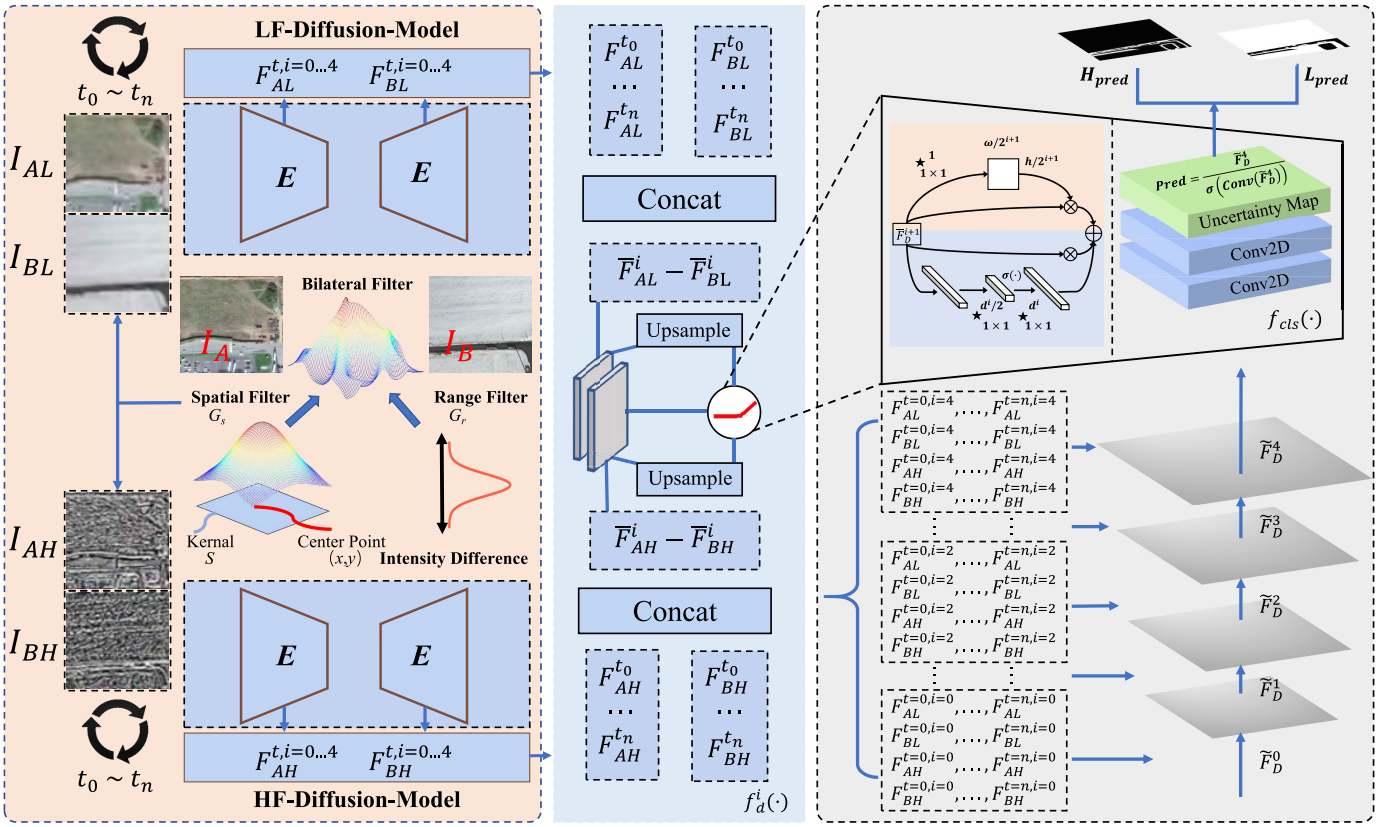


Fig. 3. Pipeline diagram about the proposed HDM-CD for RSI-CD.

comprehensive CD image, Liu et al. [23] employ a unique dual attention mechanism to attain the relationship that exists between the channel–spatial position.

To address the effects of pseudochange and noise problems regarding CD tasks, Shi et al. [24] introduce a dynamically supervised attention metric. Additionally, they have developed a unique database (referred to as SYSU-CD) aimed at overcoming the constraints faced in RSI CD. To comprehensively capture varying contextual relationships and amplify change difference features, Bandara et al. [25] introduce an RSI-CD approach using a denoising diffusion probability model (DDPM-CD). This method is the first to apply the diffusion probability model (DPM) toward development of RSI-CD, yielding promising results in detecting changes accurately. However, this approach does not fully leverage the advantages of DDPM, and fails to effectively utilize the prior information. To settle this point, Wen et al. [26] introduce a variant with inferential capabilities, termed GCD-DDPM, aiming to iteratively enhance the direct generation of precise CD maps during the inference process. Ding et al. [27] introduce a semantic change framework analog to "from-to" semantic transitions between bitime remotely sensed images, and incorporate a spatially and temporally constrained semantic learning scheme. Wen et al. [28] innovatively couple diffusion noise with conditional semantic information to propose an innovative multitype discrepancy elicitation module, overcoming the inherent issue of information loss in traditional CD algorithms, such as simple subtraction. However, these methods fail to consider frequency domain features, resulting in inaccurate localization, incomplete detection of changed objects, and incoherent boundaries. Additionally, they do not fully leverage relational changes, treating features of changed and unchanged areas indiscriminately, which limits the overall performance of existing CD methods.

III. Proposed Method

The explainable hierarchical diffusion model for RSIs change detection, namely, HDM-CD, which is presented in Fig. 3. The HDM-CD framework utilizes a hierarchical feature representation that captures both partial detail and overall structure, effectively addressing the spatial and frequency characteristics of RSIs. Additionally, it incorporates an uncertainty-guided module to focus on ambiguous regions and an SFJOM to enhance high-frequency component learning, significantly improving CD accuracy. The HDM-CD is detailed in the following section.

A. Preliminaries

1) Change Detection: In the domain of RSI-CD, the analysis typically revolves around comparing spatial data acquired at different times to discern alterations in the landscape or environment. Given two input data pairs, $\mathbf{X}_A$ and $\mathbf{X}_B$, of identical dimensions $W \times H$, both representing the same geographical region but captured at distinct time points, t_1 and t_2, respectively, using the same sensor, the scenario adheres to the conventional monomodal framework

$$\mathcal{X}_A = \{\mathcal{X}_A(x, y), 1 \le x \le W, 1 \le y \le H\} \tag{1}$$

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$$\mathcal{X}_B = \{\mathcal{X}_B(x,y), 1 \le x \le W, 1 \le y \le H\}. \quad (2)$$

The principal thrust behind the entire RSI-CD system being to yield precise binary CM

$$\mathrm{CM} = \{\mathrm{CM}(x,y) \in \{0,1\} \mid 1 \le x \le W,\ 1 \le y \le H\} \quad (3)$$

where CM is a binary matrix indicating changes between two time periods. Each element of this matrix, $\mathrm{CM}(x,y)$, can take a value of 0 or 1, where 0 signifies no change at that pixel, and 1 indicates a change. Besides, to quantify and optimize the efficacy of a CD algorithm, the learning objective for CD can be defined as follows:

$$\operatorname{argmin} \mathcal{L}(\mathrm{CM}, \mathrm{GT}) \quad (4)$$

where GT denotes the ground truth, which provides definitive information about changes. By comparing the generated CM with the ground truth, the algorithm can adjust and refine its performance, striving to reduce discrepancies and improve accuracy.

2) Diffusion Model: Diffusion probabilistic models for denoising typically employ two distinct Markov chains: one forward chain that introduces noise into the image and another reverse chain responsible for denoising the corrupted image. The forward chain represents a diffusion process that progressively injects Gaussian noise into the input data, thereby generating interference. Conversely, the reverse chain is designed to train a denoising network capable of reversing this forward diffusion. During the noise injection phase of forward diffusion, Gaussian noise is incrementally applied to the clean data $x_0 \sim p(x_0)$, leading to a complete degradation of the data until it approximates a Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. Mathematically, the forward diffusion operation at each time step t is expressed below

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t \mid \sqrt{1-\beta_t} \cdot x_{t-1}, \beta_t \cdot \mathbf{I}\right) \quad (5)$$

where $\beta_t \in (0,1)$ defines the noise schedule, and $(x_0, \ldots, x_{t-1})$ denotes a Markov chain consisting of T steps. Crucially, for a given clean data sample x_0, a noisy version x_t can be generated by resampling a Gaussian vector $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and directly employing the transformation to x_0 as follows:

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t \mid x_0 \cdot \sqrt{\bar{\alpha}_t}, (1-\bar{\alpha}_t) \cdot \mathbf{I}\right) \quad (6)$$

$$x_t = x_0 \cdot \sqrt{\bar{\alpha}_t} + \epsilon_t \cdot \sqrt{1-\bar{\alpha}_t}, \quad \epsilon_t \sim \mathcal{N}(0, \mathbf{I}). \quad (7)$$

To introduce noise into x_0, (7) is utilized to convert the data into x_t at each time step $t \in \{0, 1, \ldots, T\}$. In this context, $\bar{\alpha}_t = \prod_{i=0}^{t} \alpha_i = \prod_{i=0}^{t} (1-\beta_i)$.

B. Hierarchical Feature Representation Diffusion Model

Image low-frequency information encapsulates the structure and intensity information, such as pixel contrast, morphological modifications, and manipulations in the positioning and dimensions of objects. By focusing on particular spatial attributes, it enhances sensitivity to precise locations and configurations, thereby mitigating the incidence of false change arising from cluttered backdrops, such as alterations in vegetation and shifts in shadows due to seasonal changes. Conversely, high-frequency information augments the understanding of details and overall patterns, including variations in fine-grained textures or recurrent motifs. It could facilitate the detection of nuanced alterations within the spatial domain that may be challenging to discern. When addressing complex multiscale changes and background noise, it can discern subtle vegetative texture changes caused by seasonal variations or significant shifts in large-scale structures. Motivated by the analysis above, the process of high and low frequency information can be obtained in Algorithm 1. Subsequently, DDPM [29] has been utilized for deriving feature representations out of prechange high-frequency (I_{AH}) and low-frequency (I_{AL}) images as well as postchange I_{BH} and I_{BL}.

Algorithm 1 Frequency Separation Pseudocode

Input : Input image I, Spatial standard deviation σ_s, Range standard deviation σ_r
Output: High Frequency H, Low Frequency L
for $c \in \{R, G, B\}$ **do**
  $I_c \leftarrow I[:, :, c]$;
  for each (x, y) *in* I_c **do**
    $W_p \leftarrow 0$;
    $L_p \leftarrow 0$;
    for (x', y') *in Neighborhood of* (x, y) **do**
      $G_s \leftarrow \exp\left(-\frac{(x-x')^2+(y-y')^2}{2\sigma_s^2}\right)$;
      $G_r \leftarrow \exp\left(-\frac{(I_c(x,y)-I_c(x',y'))^2}{2\sigma_r^2}\right)$;
      $W \leftarrow G_s \cdot G_r$;
      $L_p \leftarrow L_p + W \cdot I_c(x', y')$;
      $W_p \leftarrow W_p + W$;
    end
    $L(x, y, c) \leftarrow \frac{L_p}{W_p}$;
    $H(x, y, c) \leftarrow I_c(x, y) - L(x, y, c)$;
  end
end
return L, H;

To initiate feature extraction, we start by generating high- and low-frequency images corrupted by noise $(I_{AH}^t, I_{BH}^t), (I_{AL}^t, I_{BL}^t)$, where the noise variance is defined by the time step t, resembling the pretraining process of DDPM, as illustrated below

$$I_{AH}^t = \sqrt{\gamma_t} I_{AH} + \sqrt{1-\gamma_t}\epsilon \quad (8)$$

$$I_{BH}^t = \sqrt{\gamma_t} I_{BH} + \sqrt{1-\gamma_t}\epsilon \quad (9)$$

$$I_{AL}^t = \sqrt{\gamma_t} I_{AL} + \sqrt{1-\gamma_t}\epsilon \quad (10)$$

$$I_{BL}^t = \sqrt{\gamma_t} I_{BL} + \sqrt{1-\gamma_t}\epsilon \quad (11)$$

where the noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, and the noise variance associated with time step t is expressed as $1-\gamma_t = 1 - \prod_{i=1}^{t} \alpha_i$.

Next, the noisy prechange images I_{AH}^t, I_{AL}^t and the postchange images I_{BH}^t, I_{BL}^t, comprising both high- and low-frequency components, are input into the pretrained DDPM $f_{\mathrm{DDPM}}(\cdot)$ to extract multiscale features $i \in \{0, 1, \ldots, 4\}$ corresponding to time step t

$$\left\{F_{AH}^{t,i=0}, \ldots, F_{AH}^{t,i=4}\right\} = f_{\mathrm{DDPM}}\left(I_{AH}^t\right) \quad (12)$$

$$\left\{F_{BH}^{t,i=0}, \ldots, F_{BH}^{t,i=4}\right\} = f_{\mathrm{DDPM}}\left(I_{BH}^t\right) \quad (13)$$

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$$\left\{F_{AL}^{t,i=0},\ldots,F_{AL}^{t,i=4}\right\} = f_{\text{DDPM}}\left(I_{AL}^{t}\right) \tag{14}$$

$$\left\{F_{BL}^{t,i=0},\ldots,F_{BL}^{t,i=4}\right\} = f_{\text{DDPM}}\left(I_{BL}^{t}\right) \tag{15}$$

where $\{F_{AH}^{t,i=0},\ldots,F_{AH}^{t,i=4}\}$ and $\{F_{AL}^{t,i=0},\ldots,F_{AL}^{t,i=4}\}$ denote the multiscale features characterizing the I_{AH} and I_{AL} images in the prechange period, while $\{F_{BH}^{t,i=0},\ldots,F_{BH}^{t,i=4}\}$ and $\{F_{BL}^{t,i=0},\ldots,F_{BL}^{t,i=4}\}$ represent the multiscale features within the changed I_{BH} and I_{BL} images. Here, i indicates the scales of extracted features or levels, where $i = 0$ corresponds to the lowest spatial resolution $(h/16) \times (w/16)$. Conversely, $i = 4$ represents the highest spatial resolution, which matches the resolution of the input image $h \times w$.

Finally, it is also possible to obtain features representing multiple noisy versions of the high- and low-frequency images corresponding to the pre- and postchange images. Hypothetically, the n time step represented as $t_0,\ldots,t_n$ is assumed to serve for feature extraction. Therefore, the multiscale, multitemporal time step features derived from the prechange high- and low-frequency images are given by

$$\left\{F_{AH}^{t,i=0},\ldots,F_{AH}^{t,i=4}\right\}_{t=t_0}^{t=t_n} = f_{\text{DDPM}}\left(I_{AH}^{t}\right)\Big|_{t=t_0}^{t=t_n} \tag{16}$$

$$\left\{F_{AL}^{t,i=0},\ldots,F_{AL}^{t,i=4}\right\}_{t=t_0}^{t=t_n} = f_{\text{DDPM}}\left(I_{AL}^{t}\right)\Big|_{t=t_0}^{t=t_n} \tag{17}$$

and similarly for the postchange high- and low-frequency images

$$\left\{F_{BH}^{t,i=0},\ldots,F_{BH}^{t,i=4}\right\}_{t=t_0}^{t=t_n} = f_{\text{DDPM}}\left(I_{BH}^{t}\right)\Big|_{t=t_0}^{t=t_n} \tag{18}$$

$$\left\{F_{BL}^{t,i=0},\ldots,F_{BL}^{t,i=4}\right\}_{t=t_0}^{t=t_n} = f_{\text{DDPM}}\left(I_{BL}^{t}\right)\Big|_{t=t_0}^{t=t_n}. \tag{19}$$

The hierarchical CD classifier is fine-tuned using the high- and low-frequency features across multiple scales and time steps. These features reveal the DDPM's ability to derive hierarchical semantic representations, encompassing both high and low frequencies, from the input images. Larger scale features, exemplified by $h \times w$ (i.e., $i = 4$) and $(h/2) \times (w/2)$ (i.e., $i = 3$), allow for the identification of basic patterns such as edge texture. In contrast, for those in $(h/4) \times (w/4)$ (i.e., $i = 2$), $(h/8) \times (w/8)$ (i.e., $i = 1$), and $(h/16) \times (w/16)$ (i.e., $i = 0$), scales represent more abstract characteristics. Motivated by the DDPM-CD [25], HDM-CD facilitates the generation of varied augmented feature representations by modifying the noise level introduced as an input. Such augmented features substantially improve the efficacy of RSI-CD classifier, especially when pixel-level change labels are sparse.

C. Uncertainty-Aware CD Module

The decoder of HDM-CD consists of five CD decoding blocks labeled $f_d^i(\cdot)$. The purpose of those blocks are tasked with exploiting the change features (i) on each spatial scale from the high- and low-frequency features (t) obtained over multiple time steps. Particularly, the computation of difference feature representations at each spatial scale involves multiple time step features of the high- and low-frequency images immediately pre- and postchange, combined with the difference features derived from the immediately preceding, lower scale. By iterating through these scales, the decoder effectively integrates detailed and abstract features, enhancing the accuracy within the CD procedure.

For pre- and postchange high- and low-frequency image feature sets, feature representations associated with different noise levels $t = t_0,\ldots,t_n$ for a particular scale i are concatenated separately. The specific definitions are as follows:

$$\overline{F}_{AH}^{i} = \texttt{CAT}\left(\left[F_{AH}^{t=t_0,i},\ldots,F_{AH}^{t=t_n,i}\right],\ \texttt{dim=1}\right) \tag{20}$$

$$\overline{F}_{BH}^{i} = \texttt{CAT}\left(\left[F_{BH}^{t=t_0,i},\ldots,F_{BH}^{t=t_n,i}\right],\ \texttt{dim=1}\right) \tag{21}$$

$$\overline{F}_{AL}^{i} = \texttt{CAT}\left(\left[F_{AL}^{t=t_0,i},\ldots,F_{AL}^{t=t_n,i}\right],\ \texttt{dim=1}\right) \tag{22}$$

$$\overline{F}_{BL}^{i} = \texttt{CAT}\left(\left[F_{BL}^{t=t_0,i},\ldots,F_{BL}^{t=t_n,i}\right],\ \texttt{dim=1}\right) \tag{23}$$

where CAT signifies feature concatenation, with the batch dimension treated as the 0-th dimension. The concatenated feature representations are then processed through by two 2-D convolutional layers. The initial convolutional layer decreases the dimensionality of the embedding to its original size before concatenation, denoted as d^i. Following this, the second convolutional layer produces feature representations also of size d^i. This sequence functions as a dimensionality reduction mechanism, effectively compressing the embedding dimension from $n \times d^i$ down to d^i.

By reducing the dimensionality, the model retains the critical information from multiple noise levels while ensuring that the feature representation remains efficient for subsequent processing. Subsequently, to identify and emphasize variations within the input images, the calculation of absolute differences among feature representations is required, specifically focusing on the prechange features $\overline{F}_{AH}^{i}$, $\overline{F}_{AL}^{i}$ and the postchange features $\overline{F}_{BH}^{i}$, $\overline{F}_{BL}^{i}$. The formula is given as follows:

$$F_{DH}^{i} = \left|\overline{F}_{AH}^{i} - \overline{F}_{BH}^{i}\right|_1 \tag{24}$$

$$F_{DL}^{i} = \left|\overline{F}_{AL}^{i} - \overline{F}_{BL}^{i}\right|_1 \tag{25}$$

where F_{DH}^{i} and F_{DL}^{i} signify the difference in the representation of high- and low-frequency features. Then, the difference among the high- and low-frequency feature representations from the F_{DH}^{i+1}, F_{DL}^{i+1} and (upsampled) scale $\widetilde{F}_{DH}^{i}$, $\widetilde{F}_{DL}^{i}$ are respectively combined by summation

$$\overline{F}_{DH}^{i+1} = F_{DH}^{i+1} + U\left(\widetilde{F}_{DH}^{i}\right) \tag{26}$$

$$\overline{F}_{DL}^{i+1} = F_{DL}^{i+1} + U\left(\widetilde{F}_{DL}^{i}\right) \tag{27}$$

where $U(\cdot) = \texttt{Upsample}(\cdot, \texttt{factor} = 2)$ represents the bilinear resampling operation with a scaling factor of 2.

Additionally, we adopt a difference feature enhancement strategy inspired by DDPM-CD [25]. It dynamically recalibrates the difference features in each channel and enhances the model's ability to capture complex spatial patterns across different features. The strategy can be formulated as follows:

$$\widetilde{F}_{DH}^{i} = \texttt{cSE}\left(\overline{F}_{DH}^{i}\right) + \texttt{sSE}\left(\overline{F}_{DH}^{i}\right) \tag{28}$$

$$\widetilde{F}_{DL}^{i} = \texttt{cSE}\left(\overline{F}_{DL}^{i}\right) + \texttt{sSE}\left(\overline{F}_{DL}^{i}\right) \tag{29}$$

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where $\widetilde{F}^{i}_{DH}$ and $\widetilde{F}^{i}_{DL}$ represent the high- and low-frequency outputs from the i-th hierarchical change decoder block, respectively.

The process of hierarchical change feature extraction continues iteratively through each level until it reaches the final hierarchical change decoder block, i.e., $i = 4$, which yields final difference representations of the low-frequency feature $\widetilde{F}^{4}_{DL}$ and high-frequency feature $\widetilde{F}^{4}_{DH}$

$$\widetilde{F}^{0}_{DH} = f^{0}_{d}\left(0, \left\{F^{t,i=0}_{AH}\right\}^{t_n}_{t_0}, \left\{F^{t,i=0}_{BH}\right\}^{t_n}_{t_0}\right) \tag{30}$$

$$\widetilde{F}^{0}_{DL} = f^{0}_{d}\left(0, \left\{F^{t,i=0}_{AL}\right\}^{t_n}_{t_0}, \left\{F^{t,i=0}_{BL}\right\}^{t_n}_{t_0}\right) \tag{31}$$

$$\vdots$$

$$\widetilde{F}^{4}_{DH} = f^{4}_{d}\left(\widetilde{F}^{3}_{DH}, \left\{F^{t,i=4}_{AH}\right\}^{t_n}_{t_0}, \left\{F^{t,i=4}_{BH}\right\}^{t_n}_{t_0}\right) \tag{32}$$

$$\widetilde{F}^{4}_{DL} = f^{4}_{d}\left(\widetilde{F}^{3}_{DL}, \left\{F^{t,i=4}_{AL}\right\}^{t_n}_{t_0}, \left\{F^{t,i=4}_{BL}\right\}^{t_n}_{t_0}\right). \tag{33}$$

The uncertainty of the high- and low-frequency features is calculated using convolutional layers. The formula is as follows:

$$H_{\text{pred}} = \frac{\widetilde{F}^{4}_{DH}}{\sigma\left(\text{Conv}\left(\widetilde{F}^{4}_{DH}\right)\right)} \tag{34}$$

$$L_{\text{pred}} = \frac{\widetilde{F}^{4}_{DL}}{\sigma\left(\text{Conv}\left(\widetilde{F}^{4}_{DL}\right)\right)}. \tag{35}$$

Ultimately, the representation of the extracted change features is passed via a change classifier, represented as $f_{\text{cls}}(\cdot)$, to generate the final change probability map P_{cd}

$$P_{\text{cd}} = f_{\text{cls_}h}\left(H_{\text{pred}}\right) + f_{\text{cls_}l}\left(L_{\text{pred}}\right). \tag{36}$$

D. Spatial–Frequency Joint Optimization Module

Therefore, when designing a joint spatial–frequency loss function, considering both spatial and frequency domain information ensures that the CD algorithm can capture significant, localized detail changes as well as effectively recognize widespread, holistic changes. Particularly in handling multiscale object changes and complex backgrounds, this integrated approach can effectively differentiate real changes from false positives caused by seasonal changes, shadow movement, or vegetation color shifts. The total spatial–frequency joint optimization $\mathcal{L}_{\text{total}}$ of our proposed HDM-CD framework can be described as follows:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_S + \lambda_2 \mathcal{L}_F \tag{37}$$

where $\mathcal{L}_S$ represents the spatial domain loss, and $\mathcal{L}_F$ denotes the frequency domain loss. λ_1 and λ_2 are hyperparameters to balance the optimization factors.

1) Spatial Domain Loss: Spatial domain loss primarily focuses on the local structures and pixel-level changes in images, such as pixel contrast, shape alterations, and adjustments to the position and size of objects [30]. By concentrating on specific spatial details, it provides sensitivity to exact locations and structures, thereby reducing false positives caused by cluttered backgrounds, such as vegetation changes and shadow shifts due to seasonal variations. Inspired by [31], $\texttt{Poly}_{\text{BCE}}$ is applied to evaluate the discrepancy between the generated CD image and the real CD map. Its expression is

$$\mathcal{L}_S = \texttt{Poly}_{\text{BCE}}\left(P_{\text{cd}}, P_{\text{gt}}\right) = \mathcal{L}_{\text{bce}} + \epsilon \cdot (1 - P_t)^2 \tag{38}$$

$$\mathcal{L}_{\text{bce}} = -\left[P_{\text{gt}} \cdot \log\left(P_{\text{cd}}\right) + \left(1 - P_{\text{gt}}\right) \cdot \log\left(1 - P_{\text{cd}}\right)\right] \tag{39}$$

where P_{cd} is the predicted CD map, and P_{gt} represents the real CD map. P_t represents the predicted probability for real CD maps. The ϵ is the perturbation term, $\epsilon \in \{-1, \infty\}$.

2) Frequency Domain Loss: Frequency domain loss enhances the understanding of details and overall patterns, such as changes in fine-grained textures or periodic patterns. It aids in identifying subtle changes within the spatial domain that are difficult to observe. When addressing complex multiscale changes and background noise, frequency domain loss can discern subtle vegetative texture changes caused by seasonal variations or significant shifts in large-scale structures

$$\mathcal{L}_F = \mathcal{L}_L + \mathcal{L}_H \tag{40}$$

where $\mathcal{L}_L$ denotes the loss associated with the low frequency, and $\mathcal{L}_H$ represents the loss for the high frequency. The detailed description can be introduced as follows.

3) High-Frequency Loss: High-frequency loss concentrates on the details and sharp edges within an image, involving areas of rapid change, such as alterations in buildings, renewal of vegetation, or other sudden subtle changes. This enables the detection framework all the more sensitive toward such delicate changes, enhancing the sensitivity and detail resolution of CD. Additionally, it helps minimize false detections caused by background noise, such as shadows or variations in vegetation color, thereby improving the exactness and stability of the CD results

$$\mathcal{L}_H = \lambda_3 \texttt{MSE}\left(H_{\text{pred}}, H_{\text{gt}}\right) + \lambda_4 \texttt{Poly}_{\text{BCE}}\left(H_{\text{pred}}, H_{\text{gt}}\right) \tag{41}$$

where $\texttt{MSE}$ denotes the mean squared error loss, which effectively quantifies the discrepancies at the pixel level between the predicted result and the real label map. H_{pred} and H_{gt} indicate the predicted high-frequency component and the ground-truth high-frequency component, respectively.

4) Low-Frequency Loss: Low-frequency loss focuses on the widespread and gradual changes in an image, such as large-scale color shifts, lighting differences, or regional overall changes. It aids the model to better handle large-scale changes, thereby maintaining an overall perception of change in complex scenes. In this way, it improves the HDM-CD's ability to recognize changes over extensive areas that occur gradually, maintaining a holistic view of the changes in a complex scene

$$\mathcal{L}_L = \lambda_5 \texttt{SSIM}\left(L_{\text{pred}}, L_{\text{gt}}\right) + \lambda_6 \texttt{Poly}_{\text{BCE}}\left(L_{\text{pred}}, L_{\text{gt}}\right) \tag{42}$$

where $\texttt{SSIM}$ denotes the structural similarity loss that measures the ability of the image to maintain structural integrity and is more sensitive to global information. L_{pred} and L_{gt}

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signify the predicted low-frequency component and the low-frequency component of the GT. During the training process, all hyperparameters are dynamically adjusted to automatically balance the training of the proposed HDM-CD model using GradNorm [32]. It tunes the gradient magnitudes in accordance with the characteristics of the multitask loss function. For a comprehensive discussion, the reader is referred to [32].

IV. Experiments

A. Data Description

Comprehensive experiments are provided to evaluate the effectiveness of our method, including three widely accessible RSI-CD datasets: LEVIR-CD [33], WHU-CD [34], and DSIFN-CD [35]. Detailed descriptions of these datasets are provided below.

1) LEVIR-CD Dataset: The LEVIR-CD dataset [33] contains 637 pairs of extremely high-resolution images sourced from Google Earth, each maintaining a consistent dimension of 1024 × 1024 pixels and exhibiting a spatial resolution of 0.5 m/pixel. This dataset is primarily centered around alterations related to buildings, including construction and demolition activities. In this research, a total of 445 samples were chosen at random for training purposes, 64 samples for validation, and 128 samples for testing.

2) WHU-CD Dataset: The WHU-CD [34] comprises two subsets of aerial view images, each with a resolution of 32 507 × 15 354 and a spatial resolution of 0.075 m. In this study, we cropped the images into 7600 nonoverlapping patches of fixed size 256 × 256 and classified them randomly into training, validation, and test sets at a ratio of 6:1:3.

3) DSIFN-CD Dataset: The dataset known as DSIFN [35] serves as a comprehensive resource for CD that encompasses transformations in various categories of land cover elements, including but not limited to roads, buildings, agricultural fields, and bodies of water. In the course of our experiments, every image maintains a consistent dimension of 256 × 256 pixels, resulting in 15 780 image pairs overall. Out of these, 14 276 pairs were chosen at random for the training phase, 1312 pairs were designated for validation, and 192 pairs were reserved for testing.

B. Implementation Detail

The HDM-CD model, as proposed, is executed using PyTorch version 1.11.0. During the training phase, the multilevel feature representations are extracted by the pretrained diffusion model, serving as inputs for the CD system. The Adam optimizer is employed to fine-tune the parameters of the CD detection head, while the pretrained diffusion model's parameters are kept constant. The learning rate is set initially to 1×10^{-5}, and a decay factor of 0.1 is applied. A batch size of 4 is utilized, with input images randomly cropped to dimensions of 256 × 256 pixels. The training of the HDM-CD model spans 200 epochs, resulting in the best-performing version being saved. For testing, an Intel Xeon E5-2680 v4 CPU running at 2.40 GHz is used, together with Ubuntu 18.04 and an NVIDIA GeForce RTX 3090 GPU.

C. Evaluation Metrics

Five quality assessment metrics are used for evaluation of various RSI-CD approaches: overall accuracy (OA), precision, recall, intersection over union (IoU), and F-measure (F1). All metric values range from 0 to 1, with higher values indicating greater detection accuracy.

D. Contrasting Approaches

To validate the efficacy and predominance of HDM-CD, we selected several advanced CD methods that have been selected for comparison, including FC-EF [22], FC-Siam-Conc [22], FC-Siam-Di [22], DT-SCN [23], STANet [33], SNUNet [36], BIT [2], ChangeFormer [37], DDPM-CD [25], PSTNet [38], MFATNet [39], DSAMNet [24], EGRCNN [40], ICIF-Net [41], LightCDNet [42], CTFINet [6], HANet [43], TCIANet [44], HSSENet [45], DMINet [46], DGANet [47], CCLNet [48], CCLNet++ [48], WNet [49], CSViG [50], LRDE-Net [51], GVA-CD [52], CF-GCN [17], SSCD [53], SEIFNet [54], M-Swin [55], EATDer [56], RFormer [57], GCD-DDPM [26], SMDNet [58], USFFC-Net [59], IFNET [35], and GateFormer [60].

E. Experiment on LEVIR-CD Dataset

1) Quantitative Indicator Analysis: The quantitative indicator analysis results of our HDM-CD with other competitors with regard to Pre., Rec., F1, IoU, and OA obtained over the LEVIR-CD dataset have been presented in Table I. It is revealed that HDM-CD achieves the optimal for the performance on the LEVIR-CD dataset, obtaining the supreme score in Pre. (93.16%), F1 score (91.66%), and IoU (85.61%), and achieves the first two results in terms of OA (99.29%), respectively. Particularly, the proposed HDM-CD outperforms the GateFormer by 2%, 0.94%, and 2.59% on the LEVIR-CD dataset in terms of Pre., F1 score, and IoU, respectively. Additionally, several other CD methods, such as EGRCNN, DMINet, HSSENet, and LRDE-Net, use edge prior or difference enhancement to achieve change area detection. It is clear that our HDM-CD significantly outperforms its competitors. Notably, compared to the latest method, GateFormer, as shown in Table I, HDM-CD achieves superior results in Pre. (0.9316 versus 0.9116), F1 score (0.9166 versus 0.9072), IoU (0.8561 versus 0.8302), and OA (0.9929 versus 0.9906). At the same time, compared to the DPM-based methods, that is, DDPM-CD, and GCD-DDPM, the proposed HDM-CD is significantly better than them on Pre., Rec., F1 score, IoU, and OA ([1.77%, −0.03%, 1.14%, 2.88%, 0.23%] and [2.48%, −1.6%, 0.7%, 2.05%, 0.15%]). The above-mentioned quantitative effects attest the proposed HDM-CD's outstanding performance and effectiveness in RSI-CD.

2) Subjective Visualization Analysis: For a visual comparison, illustrations on the various methods' predicted results upon the LEVIR-CD dataset are shown in Fig. 4. Several approaches are capable of detecting the predominant changes. However, they underperform in some complex scenarios (first and second rows), such as FC-EF, DT-SCN, FC-Siam-Di, and FC-Siam-Conc. These methods struggle to maintain complete boundaries of the changed objects, as well as to preserve high

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TABLE I

QUANTITATIVE RESULTS AVERAGED OVER THE LEVIR-CD AND WHU-CD DATASET FOR VARIOUS CD METHODS. ALL VALUE SETS HAVE UNIT OF %. SUPERIOR SCORES OF PRE., REC., F1, IOU, AND OA INDICATIVE THE OPTIMAL CD PROPERTIES. THE TOP THREE OPTIMUM PREDICTIVE VALUES ARE RED, BLUE, AND BLACK

Methods	Publication	LEVIR-CD *Pre.* / *Rec.* / F1 / *IoU* / *OA*	WHU-CD *Pre.* / *Rec.* / F1 / *IoU* / *OA*
FC-EF	ICIP 2018	86.91 / 80.17 / 83.40 / 71.53 / 98.39	71.63 / 67.25 / 69.37 / 53.11 / 97.61
FC-Siam-Di	ICIP 2018	89.53 / 83.31 / 86.31 / 75.92 / 98.67	47.33 / 77.66 / 58.81 / 41.66 / 95.63
FC-Siam-Conc	ICIP 2018	91.99 / 76.77 / 83.69 / 71.96 / 98.49	60.88 / 73.58 / 66.63 / 49.95 / 97.04
DT-SCN	TGRS 2020	88.53 / 86.83 / 87.67 / 78.05 / 98.77	63.92 / 82.30 / 71.95 / 56.19 / 97.42
STANet	RS 2020	83.81 / 91.00 / 87.26 / 77.40 / 98.66	79.37 / 85.50 / 82.32 / 69.95 / 98.52
SNUNet	GRSL 2021	89.18 / 87.17 / 88.16 / 78.83 / 98.82	85.60 / 81.49 / 83.50 / 71.67 / 98.71
BIT	TGRS 2021	89.24 / 89.37 / 89.31 / 80.68 / 98.92	86.64 / 81.48 / 83.98 / 72.39 / 98.75
ChangeFormer	IGARSS 2022	92.05 / 88.80 / 90.40 / 82.48 / 99.04	88.22 / 79.86 / 88.57 / 79.49 / 99.12
DDPM-CD	Arxiv 2022	91.39 / 89.67 / 90.52 / 82.73 / 99.06	92.71 / **92.05** / 92.38 / 85.84 / **99.37**
PSTNet	JSTARS 2022	90.39 / 91.09 / 90.73 / 84.11 / 98.31	91.35 / 92.01 / 91.67 / 85.15 / 98.67
MFATNet	RS 2022	91.85 / 88.93 / 90.36 / 82.42 / 99.03	93.18 / 83.93 / 88.31 / 79.08 / 99.01
DSAMNet	TGRS 2022	81.28 / 88.68 / 84.81 / 73.63 / 98.38	71.22 / **92.28** / 80.40 / 67.22 / 98.37
EGRCNN	TGRS 2022	90.24 / **91.96** / 91.09 / 81.55 / 98.85	90.92 / 89.41 / 90.16 / 82.08 / 99.29
ICIF-Net	TGRS 2022	91.32 / 88.64 / 89.96 / 81.75 / 98.99	92.98 / 85.56 / 88.32 / 79.24 / 98.96
LightCDNet	RS 2023	91.30 / 88.00 / 89.60 / 81.20 / 99.00	92.00 / 91.00 / 91.50 / 84.30 / 99.30
CTFINet	ICASSP 2023	92.52 / 89.23 / 90.28 / 82.28 / 99.07	90.44 / 83.47 / 85.83 / 75.18 / 98.69
HANet	JSTARS 2023	91.21 / 89.36 / 90.28 / 82.27 / 99.02	88.30 / 88.01 / 88.16 / 78.82 / 99.16
TCIANet	JSTARS 2023	92.51 / 89.57 / **91.59** / 83.50 / 99.05	89.45 / 86.32 / 85.91 / 75.58 / 98.98
HSSENet	TGRS 2023	**92.63** / 90.35 / 91.48 / **84.29** / **99.14**	94.56 / 88.73 / 91.55 / 84.42 / 99.35
DMINet	TGRS 2023	92.52 / 89.95 / 90.71 / 82.99 / 99.07	93.84 / 86.25 / 88.69 / 79.68 / 98.97
DGANet	TGRS 2023	92.03 / 88.56 / 90.26 / 82.25 / 99.03	**95.59** / 86.59 / 90.87 / 83.27 / **99.37**
CCLNet	TGRS 2023	91.15 / 89.85 / 90.49 / 82.64 / 99.04	86.92 / 89.37 / 88.13 / 78.78 / 99.00
CCLNet++	TGRS 2023	92.31 / 89.88 / 91.08 / 83.62 / 99.10	89.83 / 90.78 / 90.31 / 82.32 / 99.19
WNet	TGRS 2023	91.16 / 90.18 / 90.67 / 82.93 / 99.06	92.37 / 90.15 / 91.25 / 83.91 / 99.31
DMINet	TGRS 2023	92.52 / 89.95 / 90.71 / 82.99 / 99.07	93.84 / 86.25 / 88.69 / 79.68 / 98.97
CSViG	TGRS 2023	92.35 / 90.71 / **91.52** / **84.37** / **99.14**	94.69 / 86.79 / 90.57 / 82.76 / 99.28
LRDE-Net	JSTAR 2024	88.50 / **93.17** / 90.78 / ——- / 99.04	91.57 / 90.10 / 90.83 / ——- / 99.15
GVA-CD	TGRS 2024	**92.63** / 87.88 / 90.31 / 82.51 / 99.09	**95.18** / 88.07 / 91.26 / 84.41 / 98.82
CF-GCN	TGRS 2024	91.75 / 90.18 / 90.96 / 83.41 / 99.09	**94.81** / 89.04 / 91.83 / 84.90 / **99.37**
SSCD	TGRS 2024	92.12 / 89.08 / 90.58 / 82.78 / 99.06	92.45 / 88.74 / 90.19 / 82.13 / 99.27
SEIFNet	TGRS 2024	92.49 / 89.46 / 90.95 / 83.40 / 99.09	87.01 / 85.77 / 86.39 / 76.04 / 98.90
M-Swin	TGRS 2024	92.07 / 90.05 / 91.05 / 83.57 / ——-	94.55 / 91.37 / **92.93** / **86.09** / ——-
EATDer	TGRS 2024	91.74 / 90.67 / 91.20 / ——- / 98.75	91.32 / 88.74 / 90.01 / ——- / 98.58
RFormer	TGRS 2024	92.16 / 89.77 / 90.95 / 83.30 / 99.09	93.89 / 88.65 / 91.92 / 83.82 / 99.32
GCD-DDPM	TGRS 2024	90.68 / **91.24** / 90.96 / 83.56 / **99.14**	92.79 / **92.29** / **92.54** / **86.52** / **99.39**
GateFormer	JSTAR 2024	91.16 / 90.28 / 90.72 / 83.02 / 99.06	——- / ——- / ——- / ——- / ——-
SMDNet	JSTAR 2024	**92.71** / 85.89 / 90.99 / 82.71 / **99.17**	——- / ——- / ——- / ——- / ——-
HDM-CD	——-	**93.16** / 89.64 / **91.66** / **85.61** / **99.29**	93.74 / 91.99 / **93.08** / **87.05** / **99.46**

internal compactness and sharpness of edges. While ChangeFormer and BIT perform adequately in some scenes, they still exhibit misdetections when faced with pseudochanges caused by illumination (third and fourth rows). These models tend to detect inaccurate regions of changed objects, with blurred details, indicating a lack of robustness compared to our method. In cases with multiple changed objects, existing methods frequently identify pseudochange regions and overlook finer details. In contrast, DDPM-CD and HDM-CD effectively maintain the overall structure and detect changed objects of varying sizes while preserving their details (10th and 11th columns). Both DDPM-CD and HDM-CD are capable of detecting changed objects and generating fewer pseudochange pixels (fifth and sixth rows). However, HDM-CD aligns more closely with the ground truth mask, producing CMs that maintain the complete boundaries of changed objects with high internal compactness. From the visual comparison results, it is evident that HDM-CD performs effectively across various scenarios. The above-mentioned results on the LEVIR-CD dataset demonstrate that the presented HDM-CD achieves desirable results, providing higher accuracy compared to other methods.

F. Experiment on WHU-CD Dataset

1) Quantitative Indicator Analysis: Table I yields the analysis results of selected methodologies over the WHU-CD dataset. It is noticeable from Table I that although our HDM-CD method does not yield the highest results in terms of Pre., it surpasses other competitors in the main metrics, including F1, IoU, and OA. Specifically, in terms of F1 index, the proposed HDM-CD achieves a score of 93.08%, which was 0.58% superior and higher by the third most preferable method (GCD-DDPM). In terms of IoU index, the proposed HDM-CD achieves a score of 87.05%, which is 0.61% better than the second favorite method (GCD-DDPM), and 1.41% over the DDPM-CD. In terms of OA index, the proposed HDM-CD achieves a score of 99.46%, which is 0.09% over the third favorite method (DDPM-CD), and 4.01% higher than the FC-Siam-Di method. DGANet attains the best performance in terms of Pre.. However, it also exhibits an increase in false detections as a consequence. While our method makes a better balance between Pre. and Rec., it achieves the best results across three quantitative metrics. For the F1 index, our HDM-CD scores 93.08%, which is 2.48% superior than LRDE-Net. For the OA index, our method reaches 99.46%,

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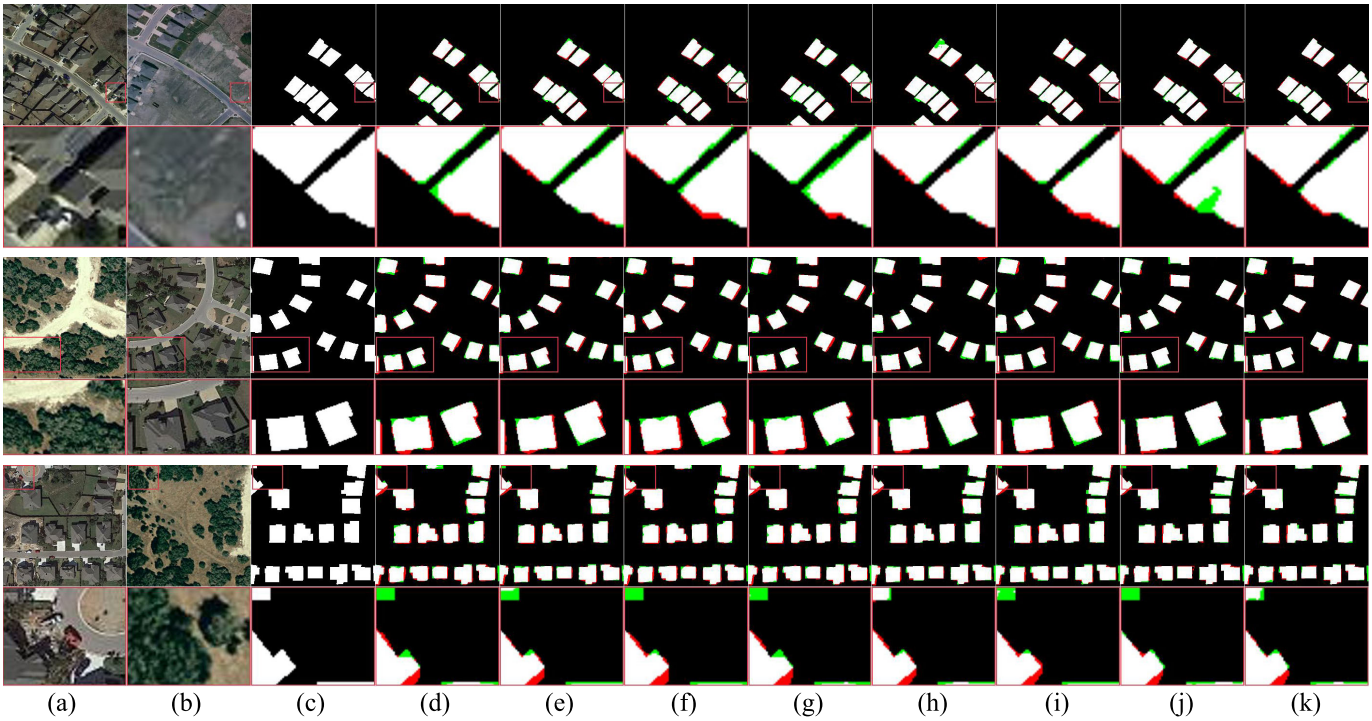


Fig. 4. Predicted results on the LEVIR-CD dataset. The second, fourth, and sixth rows display the zoomed-in view detail of the region in the red and purple rectangular areas of the first, third and fifth rows. (a) Prechange image. (b) Postchange image. (c) Ground truth mask. (d) Results of the FC-EF. (e) Results of the DT-SCN. (f) Results of the FC-Siam-Di. (g) Results of the FC-Siam-Conc. (h) Results of the ChangeFormer. (i) Results of the BIT. (j) Results of the DDPM-CD. (k) Results of the HDM-CD.

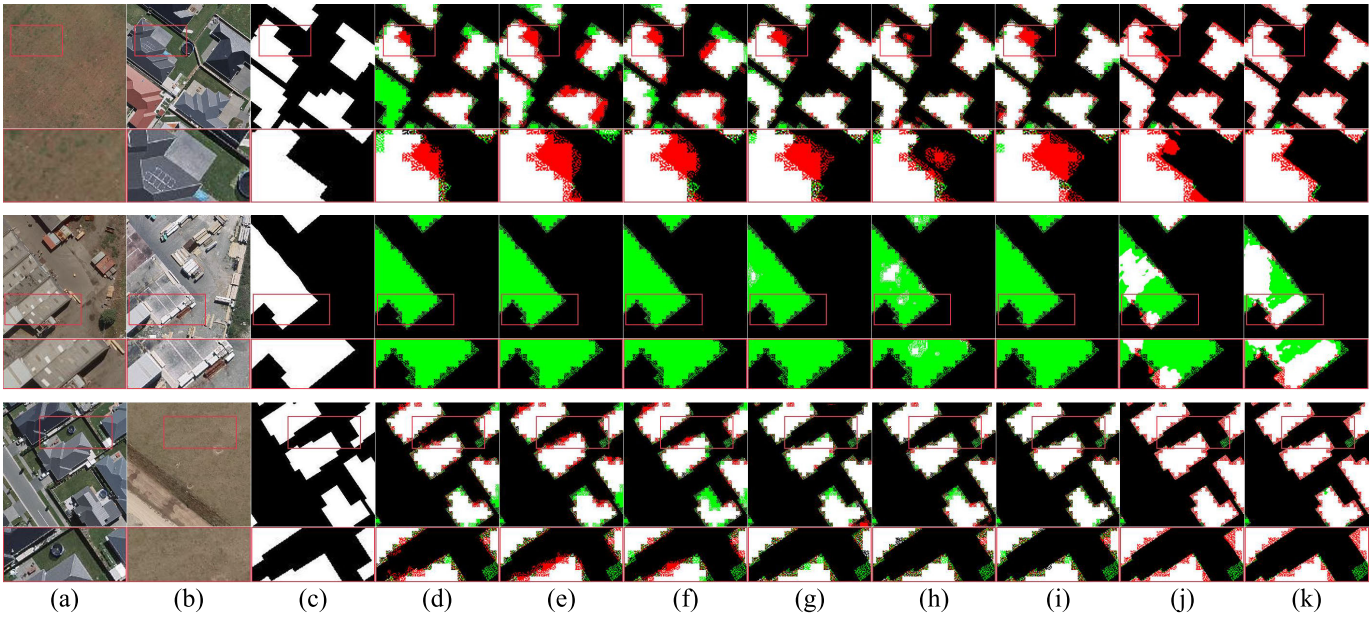


Fig. 5. Predicted results on the WHU-CD dataset. The second, fourth, and sixth rows display the zoomed-in view detail of the region in the red and purple rectangular areas of the first, third, and fifth rows. (a) Prechange image. (b) Postchange image. (c) Ground truth mask. (d) Results of the FC-EF. (e) Results of the DT-SCN. (f) Results of the FC-Siam-Di. (g) Results of the FC-Siam-Conc. (h) Results of the ChangeFormer. (i) Results of the BIT. (j) Results of the DDPM-CD. (k) Results of the HDM-CD.

surpassing the LRDE-Net method by 0.31%. Regarding the IoU index, our method attains 87.05%, which is 5.2% higher than CSViG. These quantitative results demonstrate the proposed HDM-CD's effectiveness and robustness over the WHU-CD dataset.

2) Subjective Visualization Analysis: To show the superiority of the proposed HDM-CD, we list subjective visualization analysis results of selected CD approaches, including the proposed HDM-CD in Fig. 5. Specifically selected to measure the algorithm's ability to handle regular boundaries and detect

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TABLE II

QUANTITATIVE RESULTS AVERAGED OVER THE DSIFN-CD DATASET FOR VARIOUS CD METHODS. ALL VALUE SETS HAVE UNIT OF %. SUPERIOR SCORES OF PRE., REC., F1, IOU, AND OA INDICATIVE THE OPTIMAL CD PROPERTIES. THE TOP THREE OPTIMUM PREDICTIVE VALUES ARE RED, BLUE, AND BLACK

Methods	Publication	*Pre.* / *Rec.* / F1 / *IoU* / *OA*
FC-EF	ICIP 2018	72.61 / 52.73 / 61.09 / 43.98 / 88.59
FC-Siam-Di	ICIP 2018	59.67 / 65.71 / 62.54 / 45.50 / 86.63
FC-Siam-Conc	ICIP 2018	66.45 / 54.21 / 59.71 / 42.56 / 87.57
DT-SCN	TGRS 2020	53.87 / 77.99 / 63.72 / 46.76 / 84.91
STANet	RS 2020	67.71 / 61.68 / 64.56 / 47.66 / 88.49
IFNet	ISPRS 2020	67.86 / 53.94 / 60.10 / 42.96 / 87.83
SNUNet	GRSL 2021	60.60 / 72.89 / 66.18 / 49.45 / 87.34
BIT	TGRS 2021	68.36 / 70.18 / 69.26 / 52.97 / 89.41
ChangeFormer	IGARSS 2022	89.90 / 86.01 / 87.91 / 78.43 / 95.98
PSTNet	JSTARS 2022	81.41 / 78.96 / 80.10 / 68.93 / 89.20
MFATNet	RS 2022	88.65 / 86.62 / 87.62 / 77.98 / 95.84
ICIF-Net	TGRS 2022	63.72 / 65.42 / 63.77 / 46.85 / 87.41
HSSENet	TGRS 2023	**92.52** / **89.86** / **91.17** / **83.77** / **97.04**
USFFC-Net	TGRS 2023	63.37 / 76.32 / 69.47 / —— / ——
CCLNet	TGRS 2023	85.16 / 76.85 / 80.79 / 67.78 / 93.79
CCLNet++	TGRS 2023	90.96 / 86.11 / 88.47 / 79.32 / 96.19
DMINet	TGRS 2023	60.97 / 70.92 / 65.81 / 48.75 / 87.41
CSViG	TGRS 2023	62.88 / 70.55 / 66.50 / 49.81 / 88.92
GateFormer	JSTAR 2024	**91.63** / **89.50** / **90.75** / **82.73** / **96.83**
SSCD	TGRS 2024	82.31 / 88.14 / 84.13 / 72.57 / 94.34
RFormer	TGRS 2024	67.34 / 72.19 / 69.68 / 53.47 / 89.10
SMDNet	JSTAR 2024	88.51 / 88.46 / 88.40 / 80.91 / 96.56
HDM-CD	———	**95.91** / **95.71** / **95.81** / **91.96** / **98.57**

small changed objects are regular buildings (first and fifth rows). It can be known from the results of the first row that the experimental result of FC-EF is not satisfactory, i.e., some regions of change were not detected as shown in the partial enlargement of the second column in Fig. 5(d). Approximately detecting the changes of buildings are FC-Siam-Di and FC-Siam-Conc. Approximately detecting the changes of buildings are FC-Siam-Di and FC-Siam-Conc, but the prediction map results are incomplete. The ChangeFormer, BIT, and DDPM-CD can detect large objetcs and small objects in dense buildings accurately, but the boundaries of their prediction maps are incompletely maintained in Fig. 5(h)–(j) of the second row. Obviously, the results in the third row show that most methods are unsuccessful in detecting changes in a very similar background and foreground. From the results, the proposed HDM-CD achives a relatively satisfactory predicted results in Fig. 5(k). The visual results in the fifth and sixth rows claimed that the proposed HDM-CD gives the satisfactory experimental results. It is particularly effective on fine and irregular objects, significantly reducing the missed detection of small changed objects and accurately capturing the complete boundaries of irregular objects. The above-mentioned results on the WHU-CD dataset indicate that the presented HDM-CD achieves desirable results and superior performance, demonstrating higher accuracy compared to other methods.

G. Experiment on DSIFN Dataset

1) Quantitative Indicator Analysis: The quantitative indicator analysis metrics gained from the DSIFN dataset are presented in Table II. Best, second-, and third-best results are highlighted in red, blue, and bold, respectively. As shown in Table II, our HDM-CD achieves optimal results across all indexes, demonstrating the excellent performance of the presented HDM-CD. Clearly, while FC-EF, FC-Siam-Di, and FC-Siam-Conc significantly reduce the missed detection rate of changed objects, they also incorrectly classify many changed objects as unchanged regions, resulting in low Pre. and thereby impacting the overall metrics. The main metrics, F1 score and OA, improvements ranged from 5% to 60.5% and 1.57% to 16.09%, respectively. For the IoU index, HDM-CD achieves 91.96%, which is 9.77% ahead of the suboptimal method (HSSENet) and 11.16% aware over the third-best method (GateFormer). For the Pre. index, HDM-CD reaches a score of 95.91%, which is 78.04% superior to DT-SCN. These results indicate that the presented HDM-CD effectively reduces the missed detection ratio and is capable of identifying the change region accurately. Both STANet and IFNet are severely imbalanced in Pre. and Rec., resulting in inaccurate detection performance. Likewise, CCLNet and CSViG suffer from this problem, thus resulting in low Pre., F1 score, and IoU metrics. Our HDM-CD effectively mitigates the aforementioned issues, demonstrating strong performance when handling scenarios involving several types of changed objects.

2) Subjective Visualization Analysis: To present the subjective visualization results in a more visual manner, we display the visual comparison results generated by various methods across several challenging scenarios, such as large-scale objects, small-scale objects, multiple objects, objects with minimal contrast between the foreground and background, objects with intricate details, and objects possessing internal edges. It is evident that our HDM-CD accurately highlights the correct changed object regions in these challenging cases, outperforming SOTA methods on the DSIFN dataset. Specifically, existing methods often miss details of large changed objects. In contrast, our HDM-CD not only accurately detects the changed areas of large objects but also preserves their sharp edges. Information loss is also significant for small changed objects, with some models failing to completely distinguish the background from these objects. In comparison, our HDM-CD retains more information and effectively separates small changed objects. For images with low contrast between the foreground and background, FC-EF and DT-SCN detect some areas, but the details of salient objects appear very blurred, as reflected in Fig. 6(d) and (e). For changed objects with complex details, FC-Siam-Di and FC-Siam-Conc exhibit issues such as detail loss, blurring, and unclear edges, as reflected in Fig. 6(f) and (g). The boundaries of changed objects produced by our HDM-CD are well-defined, and the predicted map closely aligns with the real label map. For changed objects with the inner edge, the BIT, ChangeForm, and DDPM-CD cannot accurately detect both semantic and nonchanged objects simultaneously. Details are reflected in Fig. 6(h)–(j). Our HDM-CD effectively preserves the internal edge information of changing objects with accurate detection of object semantic information. The qualitative and visual results across the DSIFN dataset demonstrate that our method excels in preserving both intact structure and explicit details, offering greater accuracy compared to other approaches.

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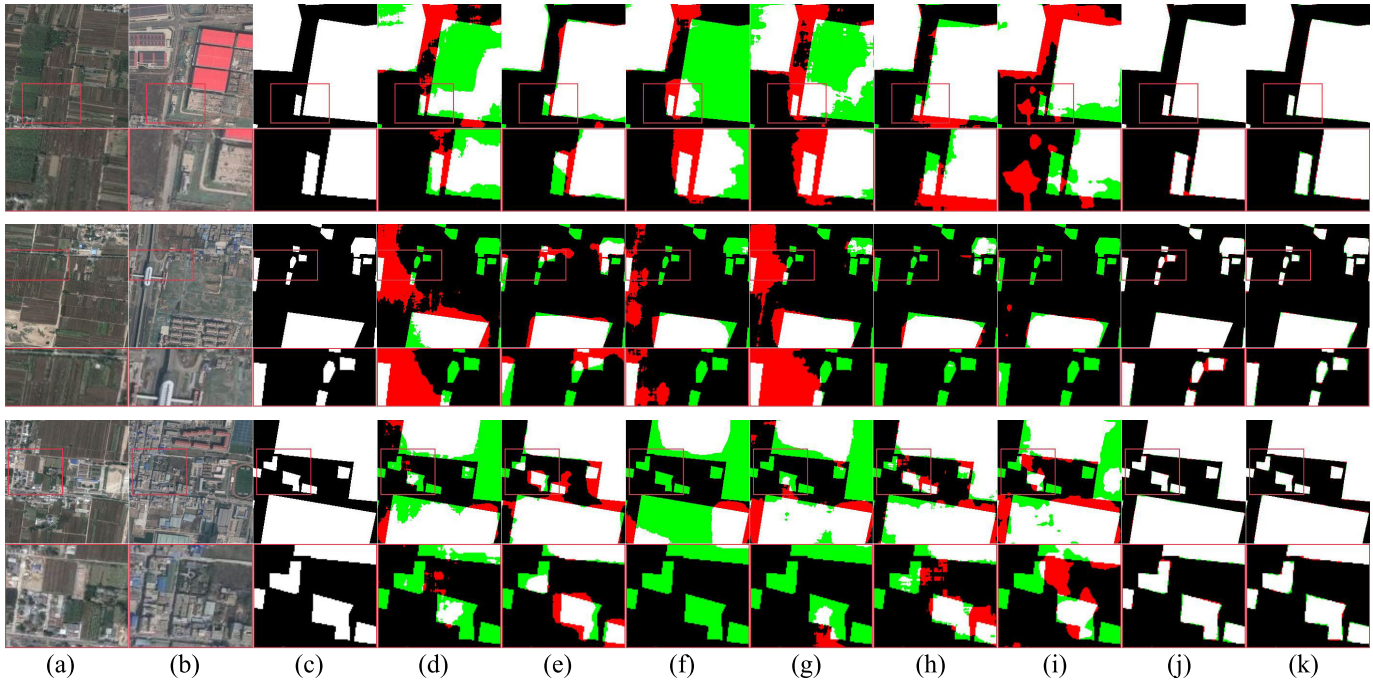


Fig. 6. Predicted results over the DSIFN-CD dataset. The second, fourth, and sixth rows display the zoomed-in view detail of the region in the red rectangular areas of the first, third, and fifth rows. (a) Prechange image. (b) Postchange image. (c) Ground truth mask. (d) Results of FC-EF. (e) Results of DT-SCN. (f) Results of FC-Siam-Di. (g) Results of FC-Siam-Conc. (h) Results of ChangeFormer. (i) Results of BIT. (j) Results of DDPM-CD. (k) Results of HDM-CD.

TABLE III
ABLATION STUDIES ANALYZED ON THE LEVIR-CD DATASET

HDM	UAM	SFJOM	*Pre.*	*Rec.*	F1	*IoU*	*OA*
Baseline			92.41	89.13	90.75	83.07	99.07
✓	✗	✗	92.72	89.17	90.91	83.34	99.09
✓	✓	✗	92.53	89.62	91.05	83.57	99.10
✓	✓	✓	**93.16**	**89.64**	**91.66**	**85.61**	**99.29**

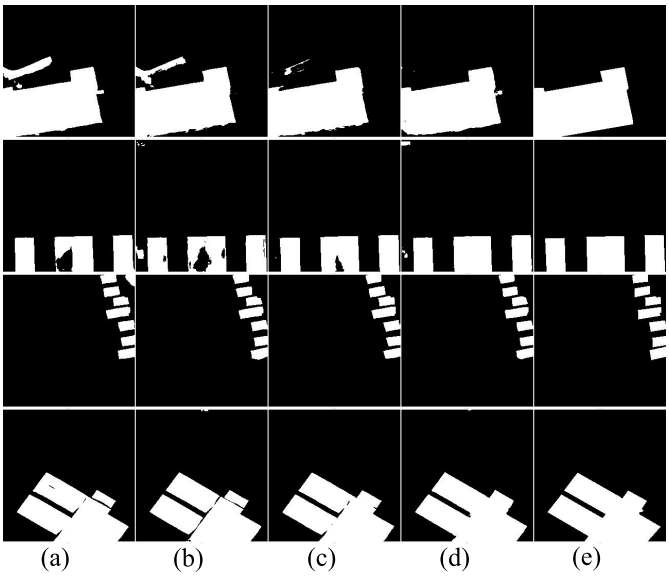


Fig. 7. Predicted results of the proposed HDM-CD with ablation studies on the LEVIR-CD dataset. (a) Baseline. (b) Baseline + HDM. (c) Baseline + HDM + UAM. (d) Baseline + HDM + UAM + SFJOM (HDM-CD). (e) Ground truth.

H. Ablation Studies

To validate the efficacy of each component, we have undertaken an ablation investigation on the LEVIR-CD dataset, systematically removing miscellaneous modules from the presented HDM-CD model. By conducting these experiments, we endeavor to present compelling evidence showcasing the usefulness of our proposed methods and to underscore their significance in improving the performance of CD algorithms.

1) Validation of HDM: RSIs frequently include objects of different scales that have experienced changes. The designed HDM can sufficiently consider the structure and texture domain information, overcoming the problem of false detection and pseudochange in the real scene. As summarized in Table III, the Pre. score improves from 0.9241 to 0.9272, the F1 value increases from 90.75% to 90.91%, and the IoU score rises from 83.07% to 83.34%. The proposed HDM shows a significant improvement over the baseline, confirming its crucial role in the experiment. It indicates that HDM effectively mitigates pseudochange information elicited by various factors, such as seasonal variations and shadow changes. As illustrated in Fig. 7(a) and (b), the baseline results display issues, for instance, misidentifying contextual areas as vital objects, or inability to accurately detect complete areas. In particular, the second row shows a notable discrepancy in the positioning of the change targets, with the change regions appearing to have blurred boundaries. However, the proposed HDM effectively addresses this issue, demonstrating its effectiveness in comparison to the baseline method.

2) Validation of UAM: By analyzing the data in Table III, it is clear that incorporating the UAM module leads to a

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TABLE IV
COMPARISON OF OUR METHOD WITH SOTA APPROACHES REGARDING PARAMETERS, TIME, AND F1 SCORE ON THE LEVIR-CD DATASET

Methods	Params (M)	Time (ms)	F1 (%)
FC-EF	1.35	24.10	83.40
FC-Siami-Di	1.35	27.71	86.31
FC-Siam-Conc	1.55	29.91	83.69
DT-SCN	41.07	37.87	87.67
STANet	16.93	40.40	87.26
SNUNet	28.34	45.67	88.16
BIT	3.55	52.69	89.31
ChangeFormer	48.63	81.40	90.40
DDPM-CD	372.93	65.58	90.52
EGRCNN	9.64	67.18	91.09
ICIF-Net	25.83	152.51	89.96
CSViG	41.31	145.38	91.52
HDM-CD	375.93	51.49	91.66

0.49% improvement in Rec. for the RSI-CD task. Additionally, there is a performance increase in several metrics: [Pre.: 0.9241 versus 0.9253], [F1: 0.9075 versus 0.9105], [IoU: 0.8307 versus 0.8357], and [OA: 0.9907 versus 0.9910]. These results confirm the efficacy of the proposed UAM module. It not only effectively filters out all kinds of pseudochange information, but at the same time is much more accurate for edge detection than other experiments in extremely complex detection contexts. The visual prediction results comparing the "baseline" model and the +UAM model are illustrated in Fig. 7(a) and (c). It is evident that the +UAM module effectively alleviates the imprecise positioning of protruding regions present in the baseline model. However, there is still room in ensuring the completeness of the identification of changing regions/objects.

3) Validation of SFJOM: To detect change regions with complex edges and irregular topology effectively, we propose SFJOM to jointly learn change regions and their edges. As observed in Table III, the main accuracy metrics exhibit an overall upward trend when the SFJOM module is incorporated. Specifically, the Rec. score improves from 0.8913 to 0.8964, the F1 value increases from 90.75% to 91.66%, and the IoU score rises from 83.07% to 85.61%. The above shows the incremental contribution of SFJOM to our approach. Likewise, as shown in Fig. 7, our method demonstrates the ability to accurately localize and fully identify change objects even against complicated backgrounds and different contrasting imagery conditions (e.g., shooting time, illumination intensity, seasonal variations, and shadow changes). Additionally, edges in the changing areas have information that is closer to the real labeling map. It is also demonstrated that SFJOM can effectively filter out validation information through inference while still reconstructing valid information about real changes. Overall, it aligns with our initial motivations to a significant extent.

I. Analysis of Complexity and Limitations

1) Analysis of Complexity: Table IV provides the computational efficiency of different models across the LEVIR-CD dataset. In particular, an intricacy analysis of our HDM-CD model has been performed alongside popular CD methods using several quantitative metrics, including the number of params, average running time (ms), and F1 score. As demonstrated in Table IV, the FC-EF, FE-Siam-Di, and FC-Siam-Conc models require minimal parameters and runtime, but this comes at the cost of their detection performance. For the F1 index, the F1 value obtained by our HDM-CD is 91.66%, which is prominent among all models. For the model parameter index, compared to the top-tier models, our network has a slightly larger model size, but the running time is lower than than ChangeFormer, EGRCNN, ICIF-Net, CSViG, and DDPM-CD. For the running time, our method takes a litter longer time than STANet and SNUNet methods, but provides more satisfactory prediction results, and the running time is much less than ChangeFormer. From the complexity assessment, it appears that the the parameters of the presented HDM-CD are reasonable, the running time is acceptable, and the detection performance is optimal.

2) Limitations: As illustrated in Table IV, although the proposed HDM-CD can produce high-resolution image CD accurately, it still has the following drawbacks: a) the sampling speed is too slow, resulting in a long model training time and b) the GPU memory usage requirement during network training is still high, resulting in a high training cost; more importantly, although our method already has a superior input resolution compared to previous CD methods, the input resolution is not infinite. For ultrahigh resolutions (e.g., 4K), the image needs to be downsampled before the input.

V. CONCLUSION

In this article, a hierarchical diffusion model called HDM-CD has been presented for the RSI-CD task, which can produce high-fidelity CD maps directly in an end-to-end fashion. It effectively harnesses spatial and frequency domain information to overcome false detection and pseudochange issues in real-world scenarios while preserving the boundaries of changing objects intact. Additionally, an uncertainty-aware module is proposed, guiding the model to prioritize the image's edge details. Furthermore, an SFJOM is devised to mitigate information loss in small object change areas and precisely detect the object contours and sharp edges within the changing regions. By comparing with numerous cutting-edge CD models exclusively across three challenging RSI-CD datasets, the presented HDM-CD model has demonstrated its superiority, establishing a leading position in the RSI-CD task.

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